

NeuroMetric estimates brain age from a single T1-weighted MRI and expresses deviation from the healthy reference trajectory as a brain age gap (PAD) in years. The estimate is decomposed into region-level contributions so every figure can be read back to the anatomy that produced it.

01 Input & preprocessing

Input is a T1-weighted volume (recommended ≤ 1.5 mm isotropic; .nii / .nii.gz / DICOM accepted). Volumes are oriented, brain-extracted and registered to a 2 mm MNI grid through a deterministic pipeline; scan-level QC flags are surfaced before inference.

02 Model

The released configuration is an ensemble of four regression models over a shared feature representation. Weights are hash-verified at load time and calibration coefficients are versioned; the calibration applied to every report is recorded on the report itself.

03 Explainability

Prediction is decomposed additively; the calibrated brain age gap is the sum of region-level contributions expressed in years and referenced to the subject's own age band. Reports therefore show *which* regions drive a deviation, not only its magnitude.

04 Reliability & protocol

Longitudinal use requires at least two visits separated by ≥ 6 months; the subject's observed brain-age slope is read against the age-normalized reference line and reported as an ageing velocity.

RELIABILITY

ICC 0.97

test-retest

PROTOCOL

2+ visits

≥ 6 months apart

GRID

MNI 2 mm

T1 · recommended input

OUTPUT

PAD + 95% CI

in years, additive

05 What the report contains

- Calibrated brain age gap (PAD) with a 95% confidence interval, in years.
- Region-level contributions in years, sorted by magnitude, positive and negative.
- Age-band referencing against the age-normalized healthy reference cohort.
- Reliability meta and provenance: model version, calibration stamp, scan identifiers.

A complete example is available as the sample report ([report.html](#)). All figures used across the site are illustrative.

06 Scope

NeuroMetric is a research-grade decision-support prototype. It is not intended for clinical diagnosis, determination or treatment decisions; a brain age gap is not, by itself, evidence of pathology. Clinical interpretation happens under physician oversight.